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ORIGINAL ARTICLE

Stock market effects of major cyber-attacks: evidence for breached and cybersecurity listed firms

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Abstract
This paper studies the short-term market reaction of major cyberattacks in the market value of breached and cybersecurity listed firms. Through an event study for 53 global cyberattacks, we find a negative and statistically significant stock price reaction for breached firms around the cyberattack disclosure dates. Our results show a higher negative stock market reaction when breached firms are small, have a higher leverage, higher growth opportunities and involves financial information loss. Finally, we also find positive and statistically significant abnormal returns for cybersecurity firms around the cyberattack disclosure date. Cyberattacks, by causing an increase in demand for security products/services, are good news for cybersecurity firms.

Keywords Cyberattack · Cyber risk · Cybersecurity firms · Market reaction · Event study

Introduction

Recent data reveals that cybercrime has inflicted losses totalling $6 trillion USD. If cybercrime were quantified as a nation, it would be the world's third-largest economy after the US and China [8]. Based on this report, many cybersecurity firms predict that the global cost of cybercrime will increase by 15 percent annually in the coming years. They expect the annual cost to reach $10.5 trillion USD by 2025. This is anticipated to be the most significant transfer of economic wealth in history, presenting risks to innovation and investment incentives. The damage caused by cybercrime seems to exceed that of natural disasters in a year. Ultimately, cybercrime is set to become more lucrative than the worldwide trade of all major illegal drugs combined [2].

The global economy is currently experiencing an unprecedented digital transformation, in a first phase triggered by COVID-19 and more recently by the adoption of artificial intelligence technologies (e.g. [19]). During this period of significant change, where any incident could potentially disrupt trust and hinder such advancement and innovations, cybersecurity is increasingly vital [22]. It is commonly acknowledged that firms fall into one of two categories: those that have experienced a breach and those that may not be aware of a breach yet [26]. With the rise in cyberspace threats, investments in cybersecurity are crucial for businesses to minimize disruptions to operations. However, given that cybersecurity behaves somewhat like a public good, leading to some firms benefiting from the cybersecurity investment made by others without contributing themselves, the equilibrium investment in cybersecurity being lower than the level that would be most beneficial for society, due to the positive impact of technical spillovers [13]. Huang et al. [21] are of the opinion that given the seriousness of the problems caused by cybercrime, firms should appoint a 'cybersecurity champion' to the board to shape the firm's approach and develop long-term strategy on cybersecurity, which according to the authors should be the priority for every organization.

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Cybersecurity threats result in direct and indirect costs for breached firms [1, 6, 7, 24]. These costs include lower sales revenue and reduced productivity due to the unavailability of breached resources, loss of funds or customer records, litigation costs, and labor and material costs needed to detect, contain, repair, and restore breached resources. Furthermore, firms that are attacked tend to see a decline in their credit ratings (e.g., [24]). This evidence suggests that firms being targeted are likely to experience a loss of reputation. Kamiya et al. [24] evidence that successful cyberattacks result in significant reputation costs, with shareholder wealth loss exceeding the attack's expenses.

Despite the high costs imposed on firms targeted by a cyberattack, empirical studies that have examined the stock price reaction to cyberattacks have found mixed results. For example, Ettredge and Richardson [11], Cavusoglu et al. [6], Arcuri et al. [1], Tosun [35], Foerderer and Schuetz [15], Hogan et al. [18], Florackis et al. [14], and Jiang et al. [23] find that cyberattacks have a statistically significant impact on stock market prices. On the contrary, Campbell et al. [5], Hovav and D'Arcy [20], and Kannan et al. [25] show that these kinds of attacks have an insignificant effect on the market values of the firms. Kamiya et al. [24] find that cyberattacks not involving the loss of personal financial information do not result in significant shareholder loss. However, if cyberattacks lead to the loss of personal financial information, shareholder wealth significantly decreases. Finally, Kvochko and Pant [26] highlight that the most recent data breaches had minimal impact on the firm's stock price, suggesting shareholders desensitization to such news.

Other empirical studies have also analyzed the effect of cyberattacks on non-breached industry peers, auditors, and affected insurers [17] and on industry competitors [24]. Both studies find that non-breached competitors suffer significant negative equity returns following the disclosure of a cybersecurity breach in their industry. Haislip et al. [17] also find that audit fees increased significantly during the year of the cybersecurity breach, and insurers with significant cybersecurity exposure experienced negative equity returns. For a more comprehensive literature, see Spanos and Angelis [34].

We contribute to the literature by examining the effect caused by the largest global cyberattacks and by extending this analysis to the twenty largest cybersecurity firms. To the best of our knowledge, this study is the first that focuses only on the impact of major cyberattacks on breached firms and simultaneously analyses its effects on the main firms producing cybersecurity tools/services (cybersecurity firms). Previous studies have analyzed the impact caused by firm type, firm size, and the year the breach occurred [6], type of information loss [24], industries, types of information, access sources and countries [18], news pressure [15] and firms' exposure to cyber risk [14, 23] in abnormal returns produced by security breaches. This study aims to fill a gap in the literature regarding the impact caused by the severity of cyberattacks, by analyzing a sample of cyberattacks with significant damage to breached firms.

Based on an event study methodology, we show that breached firms experience a significant negative and statistical stock price reaction around the cyberattack disclosure date in a sample of 53 global cyberattacks. We evidence a significant mean cumulative abnormal return (CAAR) of -4.08% over the three-day window around cyberattack announcements. This value is much higher than those found in previous empirical studies since our sample only contains cyberattacks with severe damage to breached firms. Our findings also show that when breached firms are small, have higher leverage, greater growth opportunities, and involve financial information loss, the stock market reaction is more negative. Finally, we find positive and statistically significant abnormal returns for the main cybersecurity firms. Because of the high damages caused by cyberattacks, its announcement is good news for cybersecurity firms, as there tends to be greater awareness among firms of the importance of being protected against cyberattacks, leading to a greater demand for these products/services.

Impact of cyberattacks on market value of breached and cybersecurity firms

As mentioned in the previous section, a successful cyberattack generates key financial risks to breached firms. Effects of cyberattacks on business can be financial, reputational, and legal. These effects tend to affect firm's future cashflows. The stakeholders, especially customers, tend to be afraid of transacting with the breached firm due to perceived poor security (e.g., [6], and [24]). A security breach may irreversibly harm the trust and confidence needed for a long-term relationship with customers and business partners. This can lead to the loss of customers who may switch to competitors, as well as difficulty in attracting new customers. Additionally, the perceived increase in business risk may result in higher insurance costs for the breached firm and also higher costs of capital (e.g., [24]).

A successful cyberattack often indicates to the market that the target breached firm may not prioritize customer privacy and protection of their assets and private information. It may also suggest that the firm's internal security practices are inadequate. As a result, this might cause investors to begin questioning the firm's performance [12]. When investors perceive a security breach in a negative light, believing that there will be an increase in costs and a reduction in sales revenue that will lead to a substantial decrease in expected future cash flows, breached firms may experience a negative abnormal stock market return around the time of

Table 1 Distribution of cyberattacks by country and industry

Country	#	Country	#
AUSTRALIA	2	SOUTH KOREA	1
BRAZIL	1	TURKEY	1
CHINA	2	UK	2
IRELAND	1	USA	40
JAPAN	3		
Industry	#	Industry	#
Communication services	8	Finance Services	7
Consumer cyclical—internet retail	5	Healthcare	4
Consumer cyclical—leisure, lodging, resort and casinos	5	Industrials	4
Consumer cyclical—apparel, home and auto	5	Technology	12
Consumer defensive	3		

This table presents the 53 major cyberattacks over the period 2010–2023 by country and industry

the cyberattack announcement. Therefore, our first research hypothesis is the following:

H_1: A cybersecurity attack announcement is negatively associated with abnormal stock returns for breached firms.

For cybersecurity firms, there seem to be signs that suggest a link between security breaches and their market value. A recent report from McKinsey reveals that organizations worldwide spend around $150 billion on cybersecurity, an expense that grows at a rate of 12.4% annually.[1] Previous empirical studies reveal the existence of information transfer between firms, i.e., when one group of firms makes an announcement, it significantly affects the returns of another group of non-announcing firms, providing valuable insight into interconnected market dynamics (e.g., [32]). Most of the research on information transfer has concentrated on the transfer of information within the same industry, but there are exceptions, such as the case of Olsen and Dietrich [30], who analyzed interindustry information transfers. These authors provide evidence that monthly sales announcements by retailers significantly affect the market values of both retailers and their suppliers. In the current study, the idea of information transfer is similar to Olsen and Dietrich's [30] study because it suggests that information transfer between firms that have experienced security breaches and firms that may potentially provide security products or services to these breached firms. Therefore, our second research hypothesis is the following:

H_2: A cybersecurity attack announcement is positively associated with abnormal stock returns for cybersecurity firms.

Data and methodology

The data used in this event study is gathered from diverse sources. The stock returns of firms and country stock price indices were obtained from Eikon Refinitiv. The list of major cyberattacks,[2] initially made up of 109 cyberattacks, was reduced to 53 announcements after we removed attacks on firms or organizations not publicly listed on the stock market. This list was built based on the number of users impacted, records exposed, or accounts affected. Among the firms affected are Alibaba (1.1 billion pieces of user data), Facebook (533 million users), Mariott International (500 million of customers), First American Financial Corporation (885 million file records leaked) and JP Morgan Chase (76 million households and 7 million of small businesses). The list identifies the firms targeted by the attacks, the dates when cyberattacks were disclosed to the market, and the type of information breached, for a set of cyberattacks that occurred between 2010 and 2023. Like Kamiya et al. [24], our sample includes a homogenous sample of attacks, as it only includes attacks initiated by outside parties. Specifically, we only include successful malicious external actions, such as hacking and malware. We exclude other incidents associated with internal errors or failures to follow data handling policies. Table 1 shows cyberattack distribution by country and industry. The list of the twenty largest cybersecurity firms (in terms of market capitalization) was also obtained from Eikon Refinitiv.[3]

For cross-sectional analyses, we use the control variables from Table 2, which provide their definitions, sources, and

[1] https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/cybersecurity/new-survey-reveals-2-trillion-dollar-market-opportunity-for-cybersecurity-technology-and-service-providers.

[2] The list can be found here: https://termly.io/resources/articles/biggest-data-breaches/.

[3] This list is confirmed on the following website link: https://finance.yahoo.com/news/20-biggest-cybersecurity-companies-world-212042926.html.

Table 2 Determinants of cumulative abnormal returns (CARs): variable definition and expected relationship

Variable	Notation	Measure	Author(s)	Expected Effect
Size	SIZE	Market capitalization in USD (natural logarithm)	Cavusoglu et al. [6], Haislip et al. [17]; Kamiya et al. [24], Tosun [35]	±
Institutional ownership	INST	Percentage of stock that are owned by institutional investors (%)	Kamiya et al. [24]	–/No
Leverage	LEV	Ratio of total liabilities to total assets (%)	Haislip et al. [17]; Kamiya et al. [24], Tosun [35]	–/No
Profitability	ROA	Ratio of operating income to total average assets (%)	Kamiya et al. [24], Tosun [35]	–/No
Growth opportunities	Tobin's Q	Ratio of market value to total average assets (%)	Haislip et al. [17]; Kamiya et al. [24], Tosun [35]	–
Financial information loss	FIL	Dummy that takes the value of one if a cyberattack involves financial information loss and zero otherwise	Kamiya et al. [24]	–

This table presents the definitions, notation, and the expected effect of explanatory variables of Eq. 4 on CARs (attacked firms). No = denotes absence of a statistically significant impact

expected effects. Control variables are based on accounting data from the year prior to the cyberattack disclosure, also obtained from Eikon Refinitiv. The Financial Information Loss (*FIL*) variable was built based on information collected on the website indicated in footnote 2.

To test the two research hypotheses presented in the previous section, we employ the standard abnormal returns technique, which is based on the market model.[4] We compute abnormal returns (ARs) using the dates of market disclosure of cyberattacks as event dates. Abnormal returns are computed by subtracting the expected returns, generated by the market model, from the observed returns of stock i on day t, as follows:

$$AR_{it} = R_{it} - E\left(R_{it}\right) \tag{1}$$

The event dates are assigned as day $t = 0$. We estimate the expected return on the event day using a window of −10 days and −40 days. Typically, as explained by Moser and Brauneis [29], event studies analyzing financial data use an estimation window of around 30–100 days. However, extending the estimation window may lead to overlapping event situations due to the close temporal proximity of some analyzed events. To prevent this from happening we use an estimating window of 30 days. The abnormal returns were computed using the daily return of the market index of the country where the breached firm is listed as the benchmark index. By cumulating the abnormal returns over a specific time interval, we calculate the cumulative abnormal returns (CARs) with the following specification:

$$CAR\left[t_1, t_2\right] = \sum_{t_1}^{t_2} AR_t \tag{2}$$

We considered four different time intervals for the CARs: [−1,1], [−1,5], [−5,5] and [−1,10]. Finally, we compute the cumulative average abnormal returns (CAARs) for each time intervals as follows:

$$CAAR\left[t_1, t_2\right] = 1/N \sum_{t_1}^{t_2} CAR_{[t1,t2]} \tag{3}$$

where, N is the sample size of the group. Descriptive statistics for CAARs (for breached and cybersecurity firms) are presented in Tables 3 and 4. Panel 2 of Table 3 presents the descriptive statistics for breached firms. These descriptive statistics reveal that the average market capitalization and ROA of the firms targeted by cyberattacks in the sample are $2558 million USD and 7.9%, respectively, and that 18.9% of the cyberattacks involved loss of financial information.

We use a cross-sectional ordinary least square (OLS) to estimate the effect of breached firm-specific characteristics on the variation of abnormal returns, with the following specified model:

$$\begin{aligned} CAR_{iw} &= \beta_0 + \beta_1 \ln\left(SIZE_{iw}\right) + \beta_2 INST_{iw} + \beta_3 LEV_{iw} \\ &+ \beta_4 ROA_{iw} + \beta_5 Tobin'sQ_{iw} + \beta_6 FIL_{iw} + \varepsilon_{iw} \end{aligned} \tag{4}$$

where CAR_i represents the cumulative abnormal returns for breached firm i for event window w. Table 2 presents the definition of independent variables. Robust standard errors are used in the estimation of Eq. (4).

[4] For a more in-depth understanding, look at MacKinlay [27] and Serra [33].

Table 3 Descriptive statistics of CAARs for attacked firms, control variables and abnormal returns tests

Panel 1: CAARs for attacked firms

Variable	Model	Mean	SD	25th perc	Median	75th perc	θ_1	τ_1
CAAR [−1, 1]	MM	−4.080%	6.10%	−5.67%	−1.94%	0.09%	0.000***	0.000***
CAAR [−1, 5]	MM	−5.690%	9.18%	−8.12%	−2.89%	−0.24%	0.001***	0.001***
CAAR [−5, 5]	MM	−4.814%	10.12%	−8.10%	−2.84%	0.46%	0.003***	0.003***
CAAR [−1, 10]	MM	−4.481%	13.17%	−6.53%	−2.70%	0.57%	0.005***	0.005***

Panel 2: Control variables descriptive statistics—attacked firms sample

SIZE	$ 2558 ml	$10,133 ml	$ 831 ml	$2111 ml	$3164 ml
INST	42.4%	27.0%	24.0%	48.0%	65.2%
LEV	44.7%	24.1%	33.6%	47.2%	69.0%
ROA	7.9%	12.9%	1.1%	7.0%	14.3%
Tobin's Q	54.3%	19.3%	39.0%	57.9%	66.6%
FIL	18.9%	39.1%	0	0	0

This table presents descriptive statistics of cumulative average abnormal returns (CAARs) for attacked firms around cyberattack announcement dates, calculated using market model (*MM*) (Panel 1) and control variables (Panel 2) and the results of abnormal returns tests. Firm-specific control variables are computed based on year-end accounting numbers prior to the cyberattack. The variables are the following: $SIZE_i$ is the market capitalization in USD (natural logarithm) of attacked firm *i*; $INST_i$ is the percentage of stock owned by institutional investors (%) for attacked firm *i*; LEV_i is the ratio of total liabilities to total assets (%) for attacked firm *i*; ROA_i is the ratio of operating income to total average assets (%) for attacked firm *i*; *Tobin's* Q_i is the ratio of market value to total average assets (%) for attacked firm *i*; FIL_i is a dummy variable that takes the value of one, if a cyberattack involves financial information loss and zero otherwise. θ_1 and τ_1 are the *p-values* of Brown and Warner [3, 4] t-test statistics and *z*-statistic for the sign test, respectively (see [33], for additional information)

*, ** and *** means statistical significance at the 10%, 5% and 1% level, respectively

Results

Abnormal returns

Table 3 presents the CAARs for breached firms around the cyberattack disclosure dates. We notice a decrease in stock prices that is both significant and negative around these event dates. The parametric and non-parametric tests[5] indicate a statistically significant level of 1% for the four-time intervals. The three-day window around cyberattack announcements resulted in a cumulative average abnormal return (CAAR) of −4.08%. This value is much higher than those found in recent empirical studies, which is explained by the fact that our sample only considers cyberattacks with serious damage to breached firms. Kamiya et al. [24] and Tosun [35] found a CAAR of −0.84% and −2.04% during the three-day window around cyberattack announcements, respectively. The negative stock market reaction can be explained by the fact that successful cyberattacks tend to cause significant financial, reputational, and legal damage to businesses, which tend to culminate in a reduction in expected cash-flows and in the breached firm's market value.

Regarding the stock market effect of cyberattacks on cybersecurity firms, Table 4 evidence that the Cumulative Average Abnormal Returns (CAARs) are positive and statistically significant. The parametric and non-parametric tests evidence statistical significance of at least 5% for the four-time intervals. The CAAR for the twenty biggest cybersecurity firms during the three-day window around the major cyberattack disclosure date is +0.447%. This result is explained by the high damage caused by cyberattacks to firms, where the disclosure of these attacks tends to constitute good news for cybersecurity firms, as they signal an increase in demand for cybersecurity products or services. For the four-time windows, we also compute a difference of mean test between the observed CAARs for the cybersecurity firms. The results show higher abnormal returns for larger cybersecurity firms and when the cyberattack involves financial information loss. Since large cybersecurity firms tend to have greater responsiveness, diversified market products, and brand-name recognition, they tend to capture most of the gains resulting from increased demand. Finally, cyberattacks that result in financial information loss tend to cause greater panic in firms and consequently a greater increase in cyber security investment.

Cross-section analysis

Finally, we analyze the cross-section impact of variables on the abnormal returns of breached firms. The results are in Table 5. The results also show that the stock market reacts

[5] Serra [33] explains in detail.

Table 4 Descriptive statistics of CAARs for cybersecurity firms and cumulative average abnormal returns differences for firm size and type of information loss

Variable	Mean	SD	25th perc	Median	75th perc	θ_1	τ_1
Panel 1: CAARs for cybersecurity firms							
CAAR [− 1, 1]	0.447%	0.656%	0.068%	0.271%	0.810%	0.043**	0.047**
CAAR [− -1, 5]	0.654%	0.572%	0.192%	0.809%	1.255%	0.038**	0.039**
CAAR [− 5, 5]	0.434%	0.768%	0.241%	0.422%	0.996%	0.047**	0.050**
CAAR [− 1, 10]	3.473%	1.904%	2.981%	3.069%	4.483%	0.007***	0.008***

Panel 2: Difference in the CAARs across different cybersecurity firm's size

	# Cybersecurity Firms		CAAR [− 1, 1]	CAAR [− 1, 5]	CAAR [− 5, 5]	CAAR [− 1, 10]
Above the sample median	10	CAAR	0.775%	1.055%	1.416%	4.384%
		θ_1	0.019**	0.027**	0.018**	0.002***
		τ_1	0.021**	0.029**	0.019**	0.003***
Below the sample median	10	CAAR	0.119%	0.253%	− 0.249%	2.561%
		θ_1	0.134	0.144	0.155	0.034**
		τ_1	0.139	0.142	0.162	0.039**
Difference		CAAR	0.656%	0.802%	1.965%	1.823%
		t-test *(p-value)*	0.021**	0.023**	0.018**	0.034**

Panel 3: Difference in the CAARs across different type of information loss

	# Cyberattacks		CAAR [− 1, 1]	CAAR [− 1, 5]	CAAR [− 5, 5]	CAAR [− 1, 10]
Financial Information Loss	10	CAAR	2.225%	4.130%	4.619%	7.262%
		θ_1	0.004***	0.003***	0.006***	0.002***
		τ_1	0.005***	0.004***	0.006***	0.003***
Other Information Loss	43	CAAR	− 0.150%	− 0.219%	− 0.713%	1.233%
		θ_1	0.354	0.401	0.323	0.094*
		τ_1	0.359	0.403	0.330	0.097*
Difference		CAAR	2.376%	4.349%	5.333%	6.028%
		t-test *(p-value)*	0.000***	0.000***	0.000***	0.000***

This table presents descriptive statistics of cumulative average abnormal returns (CAARs) for cybersecurity firms around cyberattack announcement dates, computed using market model (*MM*) (Panel 1) and the differences in terms of cybersecurity firms' CAARs across different subsamples of firm size (Panel 2) and type of information loss (Panel 3), for the following time windows: [− 1; + 1]; [− 1; + 5]; [− 5; + 5] and [− 1, + 10]. θ_1 and τ_1 are the *p-values* of Brown and Warner's [3, 4] *t*-test statistics and *z*-statistic for the sign test, respectively (see [33], for additional information).

The significance of the differences in CAARs is determined through two-sample *t*-test.

*, ** and *** means statistical significance at the 10%, 5% and 1% level, respectively.

negatively when breached firms are small, have higher leverage, higher growth opportunities, and involve financial information loss.

Large firms, with greater access to capital markets and multiple sources of income, and lower costs of capital (e.g., [9]) are more capable of withstanding negative economic impacts compared to small firms. According to Diamond and Verrecchia [9], given that large firms tend to disclose more information to the market, they are able to reduce information asymmetry and thus have greater access to capital from large investors at a lower cost of capital. An identical result was found by Cavusoglu et al. [6] for the variable *SIZE*. Kamiya et al. [24] show that following a security breach, firms often experience a decline in credit ratings and also tend to issue more debt. Due to high leverage and low credit ratings, the risk for the firm could increase. To alleviate the increased risk of financial distress and bankruptcy, breached firms can decide to accumulate cash due to its cushion effect and cut R&D expenses and future business expansion [35]. Lastly, the results also show more negative CAAR for breached firms that suffered financial information losses. A similar effect was obtained by Kamiya et al. [24].

Table 5 Cross-sectional analysis of CARs for attacked firms

	CAR [−1; 1]	CAR [−1; 5]	CAR [−5; 5]	CAR [−1; 10]
Constant	−0.047 (−0.884)	−0.022 (−0.433)	−0.152 (−0.978)	−0.014 (−0.078)
Ln(SIZE)	0.011* (1.769)	0.012* (1.704)	0.013* (1.775)	0.014* (1.839)
INST	0.027 (1.119)	0.022 (0.553)	0.013 (0.239)	−0.003 (−0.055)
LEV	−0.051** (−2.132)	−0.066** (−2.245)	−0.041* (−1.875)	−0.091** (−2.073)
ROA	−0.021 (−0.443)	−0.032 (−1.623)	−0.030 (−1.467)	−0.041 (−1.630)
Tobin's Q	−0.033** (−2.043)	−0.049** (−2.088)	−0.083** (−2.455)	−0.055** (−2.101)
FIL	−0.099*** (−3.193)	−0.126*** (−4.696)	−0.105*** (−2.967)	−0.156*** (−3.711)
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
#Obs	53	53	53	53
Adj. R^2	0.556	0.448	0.335	0.351

This table presents the cross-sectional estimates for the CARs for attacked firms around 53 major cyberattacks. The dependent variables are the bank CARs for four different time windows: [−1; +1]; [−1; +5]; [−5; +5] and [−1, +10], computed using market model (MM). The firm-specific variables are the following: $SIZE_i$ is the market capitalization in USD (natural logarithm) of attacked firm *i*; $INST_i$ is the percentage of stock owned by institutional investors (%) for attacked firm *i*; LEV_i is the ratio of total liabilities to total assets (%) for attacked firm *i*; ROA_i is the ratio of operating income to total average assets (%) for attacked firm *i*; *Tobin's* Q_i is the ratio of market value to total average assets (%) for attacked firm *i*; FIL_i is a dummy variable that takes the value of one, if a cyberattack involves financial information loss and zero otherwise

*, ** and *** means statistical significance at the 10%, 5% and 1% level, respectively

P-values reported in parentheses are based on standard errors adjusted for heteroskedasticity and clustering at the country level

#Obs. denotes the number of observations used in the estimation

Abnormal returns robustness check

We conduct an additional robustness to examine whether the market reaction to cyberattacks of breached and cybersecurity firms varies over time. There is no unanimity in the financial literature on this subject. Cavusoglu et al. [6] reveal that investors tend to react more harshly to the recent cyberattacks. In turn, Kamyia et al. [24] have concluded that the abnormal returns difference between the most recent and oldest subperiods are insignificant. In order to test whether there are statistically significant differences in terms of CAARs between the most recent and older attacks, as Kamiya et al. [24], we divided the sample into two subperiods of equal temporal dimension—2010–2016 and 2017–2023, calculating the CAARs for each of these subperiods and the respective test for CAARs mean differences. The results are presented in Table 6. We reached the same conclusion as Kamiya et al. [24], that although the abnormal returns are statistically significant for both subperiods, their differences are not statistically significant.

Concluding remarks

This paper examines the short-term market impact of 53 major global cyberattacks on the market value of breached and cybersecurity listed firms. This study finds that breached firms experience a significant negative impact on stock prices around the date of cyberattack disclosure. Our findings evidence that when breached firms are small, highly leveraged, have significant growth opportunities, and experience financial information loss, the stock market reaction is more negative. Finally, we show positive and statistically significant abnormal returns for cybersecurity firms around the cyberattack disclosure date. Cyberattacks, by causing an increase in demand for security products/services, are good news for cybersecurity firms. The abnormal returns observed for cybersecurity firms around the cyberattack disclosure date tend to be higher for larger firms and when they involve financial information loss.

Our findings have important implications for the financial sector. Information security is a very important issue in the financial sector, especially in the light of its potential

Table 6 Robustness check for time cumulative average abnormal returns differences

Panel 1: Difference in the CAARs for breached firms						
	# Cybersecurity Firms		CAAR [−1, 1]	CAAR [−1, 5]	CAAR [−5, 5]	CAAR [−1, 10]
Time period 2010–2016	15	CAAR	−3.709%	−5.287%	−4.556%	−4.041%
		θ_1	0.006***	0.008***	0.007***	0.009***
		τ_1	0.007***	0.009***	0.007***	0.010***
Time period 2017–2023	38	CAAR	−4.326%	−6.085%	−5.011%	−4.779%
		θ_1	0.000***	0.000***	0.002***	0.004***
		τ_1	0.000***	0.000***	0.003***	0.005***
Difference		CAAR	0.617%	0.798%	0.455%	0.738%
		t-test *(p-value)*	0.134	0.141	0.194	0.188
Panel 2: Difference in the CAARs for cybersecurity firms						
	# Cyberattacks		CAAR [−1, 1]	CAAR [−1, 5]	CAAR [−5, 5]	CAAR [−1, 10]
Time period 2010–2016	15	CAAR	0.408%	0.597%	0.394%	3.212%
		θ_1	0.053*	0.046**	0.066*	0.012**
		τ_1	0.055*	0.048**	0.064*	0.014**
Time period 2017–2023	38	CAAR	0.467%	0.687%	0.467%	3.698%
		θ_1	0.037**	0.035**	0.042**	0.006***
		τ_1	0.039**	0.036**	0.044**	0.005***
Difference		CAAR	−0.059%	−0.090%	−0.073%	−0.486%
		t-test *(p-value)*	0.568	0.544	0.598	0.133

This table presents the differences for breached (Panel 1) and cybersecurity firms' (Panel 2) in terms of CAARs across two subsamples of time—2010–2016 *versus* 2017–2023, for the following time windows: [−1; +1]; [−1;+5]; [−5;+5] and [−1,+10]. θ_1 and τ_1 are the *p-values* of Brown and Warner's [3, 4] *t*-test statistics and *z*-statistic for the sign test, respectively (see [33], for additional information)

The significance of the differences in CAARs is determined through two-sample *t*-test

*, ** and *** means statistical significance at the 10%, 5% and 1% level, respectively

impact on reputation (e.g., [24]). For financial intermediaries' reputation is crucial, considering that the supply of payment, risk management services and asymmetric information all create systemic risk (e.g., [16]). Our results, by showing that firms that suffer cyberattacks that result in loss of financial information are the most penalized by the market, highlight the importance of cyber risk management in the financial sector. In this regard Eisenbach et al. [10] show that a cyberattack on a major bank could significantly disrupt the financial system, affecting the solvency of many other banks, particularly banks with less liquidity or more unstable funding. Given the growing and significant online presence of banking industry, an innovative and proactive approach to risk management is essential as banks move into this new territory [31]. From what has been explained, cyber risk should be an important category of banking risk. It is extremely important that boards, especially of financial institutions, consider this type of risk.

Although our study offers valuable insights, it is important to consider its limitations. The recent study by Hogan et al. [18] reveals the existence of different impacts per country. In the case of the present sample, it is expected that there will be different abnormal returns for emerging countries given the differences in terms of regulation, property rights protection and market efficiency (e.g., [28]). However, given the small number of emerging countries in the sample, this hypothesis cannot be tested in the present study. Therefore, the results presented in the study should be read with caution.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Arcuri, M. C., M. Brogi, and G. Gandolfi. 2018. The effect of cyber-attacks on stock returns. *Corporate Ownership & Control* 15 (2): 70–83.
2. Broadhurst, R. 2017. *Cybercrime: Thieves, Swindlers, Bandits, and Privateers in Cyberspace*. In The Oxford Handbook of Cyber Security. Oxford, UK: Oxford Handbooks Press.
3. Brown, S. J., and J. B. Warner. 1980. Measuring security price performance. *Journal of Financial Economics* 8 (3): 205–258.
4. Brown, S. J., and J. B. Warner. 1985. Using daily stock returns: The case of event studies. *Journal Of Financial Economics* 14 (1): 3–31.
5. Campbell, K., L. A. Gordon, M. P. Loeb, and L. Zhou. 2003. The economic cost of publicly announced information security breaches: Empirical evidence from the stock market. *Journal of Computer Security* 11 (3): 431–448.
6. Cavusoglu, H., B. Mishra, and S. Raghunathan. 2004. The effect of internet security breach announcements on market value: Capital market reactions for breached firms and internet security developers. *International Journal of Electronic Commerce* 9 (1): 70–104.
7. Corbet, S., and C. Gurdgiev. 2019. What the hack: Systematic risk contagion from cyber events. *International Review of Financial Analysis* 65:101386.
8. Cybercrime Magazine. 2020. Cybercrime to Cost the World $10.5 Trillion Annually by 2025. Available here: https://cybersecurityventures.com/hackerpocalypse-cybercrime-report-2016/. Accessed on May 16, 2024.
9. Diamond, D. W., and R. E. Verrecchia. 1991. Disclosure, liquidity, and the cost of capital. *The Journal of Finance* 46 (4): 1325–1359.
10. Eisenbach, T. M., A. Kovner, and M. J. Lee. 2022. Cyber risk and the US financial system: A pre-mortem analysis. *Journal Of Financial Economics* 145 (3): 802–826.
11. Ettredge, M. L., and V. J. Richardson. 2003. Information transfer among internet firms: The case of hacker attacks. *Journal of Information Systems* 17 (2): 71–82.
12. Fama, E. F., L. Fisher, M. C. Jensen, and R. Roll. 1969. The adjustment of stock prices to new information. *International Economic Review* 10 (1): 1–21.
13. Fedele, A., and C. Roner. 2022. Dangerous games: A literature review on cybersecurity investments. *Journal Of Economic Surveys* 36 (1): 157–187.
14. Florackis, C., C. Louca, R. Michaely, and M. Weber. 2023. Cybersecurity risk. *The Review Of Financial Studies* 36 (1): 351–407.
15. Foerderer, J., and S. W. Schuetz. 2022. Data breach announcements and stock market reactions: A matter of timing? *Management Science* 68 (10): 7298–7322.
16. Gillet, R., G. Hübner, and S. Plunus. 2010. Operational risk and reputation in the financial industry. *Journal of Banking & Finance* 34 (1): 224–235.
17. Haislip, J., K. Kolev, R. Pinsker, and T. Steffen. 2019. The economic cost of cybersecurity breaches: A broad-based analysis. *In Workshop on the Economics of Information Security* (WEIS), pp. 1–37.
18. Hogan, K. M., G. T. Olson, J. D. Mills, and P. A. Zaleski. 2023. An analysis of cyber breaches and effects on shareholder wealth. *International Journal of the Economics of Business* 30 (1): 51–78.
19. Holmström, J. 2022. From AI to digital transformation: The AI readiness framework. *Business Horizons* 65 (3): 329–339.
20. Hovav, A., and J. D'Arcy. 2003. The impact of denial-of-service attack announcements on the market value of firms. *Risk Management and Insurance Review* 6 (2): 97–121.
21. Huang, K., X. Wang, W. Wei, and S. Madnick. 2023. The Devastating Business Impacts of a Cyber Breach. *Harvard Business Review* 1–6.
22. International Monetary Fund. 2021. The global cyber threat. *Finance and Development*. Available here: https://www.imf.org/external/pubs/ft/fandd/2021/03/global-cyber-threat-to-financial-systems-maurer.htm. Accessed on May 16, 2024.
23. Jiang, H., N. Khanna, Q. Yang, and J. Zhou. 2024. The cyber risk premium. *Management Science* 70 (12): 8791–8817.
24. Kamiya, S., J. K. Kang, J. Kim, A. Milidonis, and R. M. Stulz. 2021. Risk management, firm reputation, and the impact of successful cyberattacks on target firms. *Journal of Financial Economics* 139 (3): 719–749.
25. Kannan, K., J. Rees, and S. Sridhar. 2007. Market reactions to information security breach announcements: An empirical analysis. *International Journal of Electronic Commerce* 12 (1): 69–91.
26. Kvochko, E., and R. Pant. 2015. Why data breaches don't hurt stock prices. *Harvard Business Review* 31:2015.
27. MacKinlay, A. C. 1997. Event studies in economics and finance. *Journal of Economic Literature* 35 (1): 13–39.
28. Morck, R., B. Yeung, and W. Yu. 2000. The information content of stock markets: Why do emerging markets have synchronous stock price movements? *Journal of Financial Economics* 58 (1–2): 215–260.
29. Moser, S., and A. Brauneis. 2023. Should you listen to crypto YouTubers? *Finance Research Letters* 54:103782.
30. Olsen, C., and J. R. Dietrich. 1985. Vertical information transfers: The association between retailers' sales announcements and suppliers' security returns. *Journal of Accounting Research*. https://doi.org/10.2307/2490695.
31. Pennathur, A. K. 2001. "Clicks and Bricks": E-risk management for banks in the age of the internet. *Journal of Banking & Finance* 25 (11): 2103–2123.
32. Schipper, K. 1990. Information transfers. *Accounting Horizons* 4 (4): 97.
33. Serra, A. P. 2004. Event study tests: A brief survey. *Gestão. Org-Revista Electrónica de Gestão Organizacional* 2 (3): 248–255.
34. Spanos, G., and L. Angelis. 2016. The impact of information security events to the stock market: A systematic literature review. *Computers & Security* 58:216–229.
35. Tosun, O. K. 2021. Cyber-attacks and stock market activity. *International Review of Financial Analysis* 76:101795.

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